

Label-Efficient Dark-Vessel Detection in SAR: Does SAR-Domain Pretraining Outperform Optical and ImageNet Transfer Across ViT and CNN?

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Abstract. Detecting dark vessels—vessels lacking matched Automatic Identification System (AIS) reports—in spaceborne synthetic aperture radar (SAR) plays an important role in supporting maritime safety and enforcement, including monitoring activity potentially associated with sanctions evasion. Since human-verified examples are scarce, we study whether source-domain pretraining improves label efficiency on xView3 (Sentinel-1 GRD, 10 m pixel spacing) and whether the effect persists across architectures. Our controlled comparison uses approximately size-matched ViT-B/16 (86M parameters) and ConvNeXt-V2-Base (89M) tracks. Within each track, we compare random initialization with pre-trained encoders from ImageNet, optical remote sensing, and SAR. The ViT track uses AugReg, SatDINO (DINO on fMoW-RGB), and SAR-MAE (masked autoencoding on SAR-1M); the ConvNeXt track uses FC-MAE and BigEarthNet Sentinel-2 and Sentinel-1 models. Because the checkpoints differ in pretraining recipe as well as source domain, domain effects are interpreted within architecture. Matched-role cross-track comparisons test whether the observed pattern persists across architectures. All arms use the same [VH, VV, VH–VV] input, CenterNet-style, anchor-free Gaussian-heatmap point detector, optimizer, schedule, scene-level splits, and seed. We fine-tune nested 10%, 25%, 50%, and 100% fractions of the training scenes in a stratified 150-scene study subset. We report seed-0 distance-matched F1 on held-out scenes, plus dark-vessel recall and near-shore F1 on a disjoint, near-shore, 50-scene human-verified set used once for final evaluation. Because training subsets are not restricted to near-shore scenes, this final measure tests coastal generalization under distinct clutter conditions. All reported values are seed-0 point estimates.

Keywords: Synthetic aperture radar · Dark-vessel detection · Transfer learning · Label efficiency · Vision transformers · Convolutional neural networks · xView3-SAR.